Different research methods/designs

- Action Research
- Experimental Research
- Case study approach
- Surveys
- Historical Research

Action Research

- Action research has been used particularly by professionals who want to investigate and improve their own working practices.
- For example,
  - nurses might enquire into their interactions with patients and try different approaches to improving their work
  - Teachers apply different strategies for maintaining discipline in the classroom and examine which seem to be the most effective.

- Action research is categorized by :
  - Concentration or practical issues:
    - Requires researchers to work in the field, in the messi-ness of human affairs.
  - An iterative cycle of plan-act-reflect:
    - the researchers plan to do something in a real-world situation, do it, and then reflect on what happened or was learnt, and then begin another cycle of plan—act-reflect.
  - An emphasis on change:
    - researchers do not simply observe and describe. They are concerned with doing things that make a difference, and learning about how they effected the change.

- Collaboration with practitioners:
  - people living and working in the situation under study are active participants in the research.
- Multiple data generation methods:
  - there are no restrictions on the types of data appropriate to action research. Both quantitative and qualitative data can be used. Researchers might decide to interview and observe, issue questionnaires, and collect documents produced by themselves (for example, their own field notes or personal journals) or by others in the situation (for example, minutes of departmental meetings, web-server logs or internal briefing notes).

– Action outcomes plus research outcomes:
    - action research outcomes can relate to both action (practical achievements in the problem situation) and research (learning about the processes of problem-solving and acting in a situation).

- Examples of action research in CS:
  - Action research in CS is practical, context-specific, and aims at continuous improvement in real-world settings (classroom, organization, software team, lab, etc.) rather than controlled lab-only experiments**.**
  - Topics that fit into the cyclical process of planning → acting → observing → reflecting → refining
- **Improving Student Learning in Programming Courses**
  - **Problem**: Students struggle to grasp recursion concepts in introductory programming.
  - **Action**: The teacher introduces visualization tools and problem-based learning activities.
  - **Observation**: Collects student performance data, quiz results, and feedback.
  - **Reflection**: Analyzes which visualization methods worked best and refines the teaching strategy for the next batch.

- **Enhancing Software Development Practices in a Team**
  - **Problem**: A software development team experiences frequent bugs in production.
  - **Action**: Introduce pair programming and continuous integration practices.
  - **Observation**: Monitor defect rates, development speed, and team collaboration.
  - **Reflection**: Assess effectiveness, retain useful practices, and adjust workflow further.

- **Improving Cybersecurity Awareness in an Organization**
  - **Problem**: Employees often fall for phishing emails.
  - **Action**: Conduct interactive workshops and simulate phishing attacks.
  - **Observation**: Track how many employees still click on malicious links before/after the intervention.
  - **Reflection**: Adapt training strategies (e.g., gamification, reminders) and re-test.

- **Optimizing Database Query Performance in a University Project**
  - **Problem**: Student project databases are slow due to poorly written SQL queries.
  - **Action**: Introduce query optimization guidelines and query tuning workshops.
  - **Observation**: Measure query execution times before and after optimization.
  - **Reflection**: Refine guidelines and share best practices with next year’s students.

- **Reducing Energy Consumption in Computer Labs**
  - **Problem**: Computers in labs are left powered on, wasting electricity.
  - **Action**: Deploy automated power management scripts (shutdown/hibernation after inactivity).
  - **Observation**: Monitor energy consumption before and after implementation.
  - **Reflection**: Improve automation policies and scale up across labs.

- **Adopting AI Tools for Code Review in Academia**
  - **Problem**: Manual code review in student projects is time-consuming and inconsistent.
  - **Action**: Introduce an AI-based code review assistant to help in grading and feedback.
  - **Observation**: Collect data on grading time, student learning outcomes, and feedback quality.
  - **Reflection**: Adjust tool usage policies and provide training for both students and instructors.

Case Studies

- A case study focuses on one instance of the ‘thing’ that is to be investigated: an organi- zation, a department, an information system, a discussion forum, a systems developer, a in development project, a decision, and so on.
- This one instance, or case, is studied depth, using a variety of data generation methods (interviewing, observation, document analysis and/or questionnaires).
- The aim is to obtain a rich, detailed insight into the ‘life’ of that case and its complex relationships and processes.

- A case study is characterized by:
  - **Focus on depth rather than breadth:** the researcher obtains as much detail as possible about one instance of the phenomenon under investigation.
  - **Natural setting:** the instance, or case, is examined in its natural setting, not in a laboratory or other artificial situation. The case existed prior to the researcher arriving on the scene, and, normally, continues to exist after the researcher has moved on. The researcher seeks to disturb the setting as little as possible.
  - **Holistic study:** the researcher focuses on the complexity of relationships and processes and how they are interconnected and inter-related, rather than trying to isolate individual factors.
  - **Multiple sources and methods:** the researcher uses a wide range of data source, For example, if studying a department, a researcher will try to talk to as many people as possible about life and work in that department, rather than just one or two people, to obtain multiple perceptions about the department and how it operates

- Examples in Computer Science
  - Case study research in **Computer Science** typically investigates a **single case (or a small set of cases) in depth**, often in a real-world context.
  - Unlike action research, case studies are more **observational and explanatory** than intervention-based.
- The difference from action research:
  - **Case Study** = In-depth *analysis* of an existing system/project in its natural setting.
  - **Action Research** = Involves *intervention* and iterative improvement.

- **Adoption of Cloud Computing in a University**
  - **Case**: A university migrating its learning management system (LMS) from on-premises servers to a cloud platform.
  - **Focus**: Challenges in cost, scalability, security, and user experience.
  - **Research Outcome**: Insights into benefits, risks, and best practices for educational institutions moving to the cloud.

- **Agile Methodology in Software Development Teams**
  - **Case**: A mid-sized IT company transitioning from Waterfall to Agile.
  - **Focus**: Impact on productivity, software quality, and developer satisfaction.
  - **Research Outcome**: Lessons learned, difficulties faced, and guidelines for smoother adoption of Agile in similar companies.

- **Use of AI-Based Code Review Tools in Academia**
  - **Case**: A computer science department integrating automated AI tools to evaluate student coding assignments.
  - **Focus**: Accuracy, fairness, time saved, and student perceptions of feedback.
  - **Research Outcome**: Practical considerations and ethical issues in deploying AI in education.

- **Case Study of E-Governance Portal Implementation**
  - **Case**: A state government developing a digital platform for citizen services.
  - **Focus**: Usability, scalability, cybersecurity, and citizen adoption.
  - **Research Outcome**: Policy recommendations and technology integration strategies

- **Blockchain in Supply Chain Management**
  - **Case**: A company adopting blockchain for tracking goods and ensuring authenticity.
  - **Focus**: Technical challenges, scalability, cost, and stakeholder trust.
  - **Research Outcome**: Evidence-based evaluation of blockchain's practicality in supply chains.

- **Case Study of Smart City IoT Implementation**
  - **Case**: Deployment of IoT-enabled traffic management in a metropolitan area.
  - **Focus**: Network architecture, data privacy, scalability, and real-time analytics.
  - **Research Outcome**: Guidelines for IoT implementation in other smart city projects.

Experimental research

- In academic research, an experiment is a strategy that investigates cause and effect relationships, seeking to prove or disprove a causal link between a factor and an observed outcome.
- Researchers start by developing a theory about their topic of interest, which leads to a statement based on the theory that can be B, and tested empirically via an experiment.
- This statement is of the form ‘Factor A causes and is known as a hypothesis.
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- A research strategy based on experiments is typically characterized by:
  - Observation and measurement.
  - process of (1) observation or measurement of a factor; (2) manipulation of circumstances; and (3) re-observation or re-measurement of the factor to identify any changes.
  - Identification of causal factors.
  - Explanation and prediction.
  - Repetition. Experiments are typically repeated many times, and under varying conditions, to be certain that the observed/measured outcomes are not caused by some other factor.

- In **Computer Science**, **experimental research** is when you manipulate variables in a controlled environment to test hypotheses and measure effects.
- Key features of **Experimental Research in CS**:
  - Controlled setting (labs, simulators, testbeds, or real-world controlled trials).
  - Clear **independent (what you change)** and **dependent (what you measure)** variables.
  - Often repeatable and quantifiable.

- **Algorithm Performance Testing**
  - **Experiment**: Compare different sorting algorithms (QuickSort, MergeSort, HeapSort) under various input sizes and distributions.
  - **Independent Variable**: Sorting algorithm used.
  - **Dependent Variable**: Execution time, memory usage.
  - **Outcome**: Identify which algorithm performs best under specific conditions.

- **Evaluating Machine Learning Models**
  - **Experiment**: Train multiple ML classifiers (SVM, Random Forest, Neural Network) on a dataset for fraud detection.
  - **Independent Variable**: Type of algorithm.
  - **Dependent Variable**: Accuracy, precision, recall, F1-score.
  - **Outcome**: Evidence for choosing the best algorithm for fraud detection.

- **Impact of User Interface (UI) Design on Usability**
  - **Experiment**: Test two versions of a mobile app (minimalist vs. feature-rich interface).
  - **Independent Variable**: Interface design.
  - **Dependent Variable**: Task completion time, error rate, user satisfaction.
  - **Outcome**: Determine which design is more user-friendly.

- **Performance of Networking Protocols**
  - **Experiment**: Compare TCP, UDP, and QUIC protocols under varying network loads.
  - **Independent Variable**: Protocol type.
  - **Dependent Variable**: Latency, throughput, packet loss.
  - **Outcome**: Decide which protocol is most effective for high-speed communication.

Surveys

- Survey research usually involves **collecting opinions, perceptions, or experiences** from a group of respondents using questionnaires, interviews, or online forms.
- **Survey research in Computer Science often focuses on perceptions, usage patterns, and attitudes** of students, developers, or users about technologies, tools, or practices.

- **Online Learning Platforms Usage Survey**
  - Research Objective: To study how postgraduate computer science students perceive the effectiveness of platforms like Coursera, Udemy, and edX.
  - Method: Online questionnaire on frequency of use, ease of learning, and satisfaction.
  - Outcome: Identify factors influencing adoption of e-learning in CS education.

- **Cyber security Awareness among College Students**
  - Research Objective: To assess the level of awareness about phishing, password security, and safe browsing among undergraduates.
  - Method: A structured survey distributed to computer science students.
  - Outcome: Suggests improvements in cybersecurity training curriculum.

- **Survey on Developers’ Preferences for Programming Languages**
  - Research Objective: To identify which programming languages developers prefer for web, AI, and mobile app development.
  - Method: Online survey targeting professional developers via LinkedIn or GitHub communities.
  - Outcome: Understanding industry trends in programming language adoption.

- **Usability Survey of Mobile Banking Apps**
  - Research Objective: To evaluate user experience of mobile banking apps in terms of performance, UI/UX, and security.
  - Method: Likert-scale based survey with app users.
  - Outcome: Helps improve app usability features.

- **Impact of Social Media on Academic Performance of Computer Science Students**
  - Research Objective: To understand how social media usage affects concentration, coding practice, and research productivity.
  - Method: Survey with time-tracking and self-reported performance data.
  - Outcome: Insights into balancing social media and academic life.

Historical research

- Historical research involves **studying past events, developments, or trends** in computer science to understand how ideas, technologies, or practices evolved over time.
- It usually relies on **archival records, old publications, system documentation, patents, interviews, or case histories**.
- In short: **Historical research in computer science** looks at **where technologies came from, how they developed, and what lessons they provide for today's innovations**.

- **History of Artificial Intelligence**
  - Tracing AI from the **1956 Dartmouth Conference**, through **expert systems in the 1980s**, to **modern machine learning and generative AI**.
  - Focus: How shifts in hardware, algorithms, and data availability shaped AI.

- **Evolution of Programming Languages**
  - Study of how programming languages progressed from **Assembly** → **Fortran** → **C** → **Java** → **Python** → **AI-driven languages**.
  - Focus: Why new languages emerged, what problems they solved, and their adoption in academia/industry.

- **History of Human–Computer Interaction (HCI)**
  - Studying the evolution from **command-line interfaces** → **GUIs** → **touchscreens** → **AR/VR** → **brain-computer interfaces**.
  - Focus: How usability research and human factors shaped design.

- **Women in Computing – Historical Contributions**
  - Documenting contributions of **Ada Lovelace, Grace Hopper, Margaret Hamilton, Radia Perlman**, and others.
  - Focus: Understanding gender roles and recognition in computing history.